PH4207

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Assignment - 1

1(a) The Gas is isothermally compressed V_i to V_f. Before compression a molecule can be anywhere in V_i, after compression it is restricted to V_f. So, probability of finding the molecules in smaller ~~volume~~ volumn is related to $\frac{V_i}{V_f}$.

Information per molecules $\rightarrow$ $I_1 = -\log_2\left(\frac{V_f}{V_i}\right)$

$= \log_2\left(\frac{V_i}{V_f}\right)$

Total information $\rightarrow$ $N I_1 = N \log_2\left(\frac{V_i}{V_f}\right)$

Now, entropy variation $= \Delta S$

$$\Delta Q = -\int_{V_i}^{V_f} P\,dV = -\int_{V_i}^{V_f} \frac{N k_B T}{V}\,dV = -N k_B T \ln\frac{V_f}{V_i} = N k_B T \ln\frac{V_i}{V_f}$$

Now we can write

$$\Delta S = N k_B T \ln\frac{V_i}{V_f}$$

$$= N k_B \frac{\ln\frac{V_i}{V_f}}{\ln 2} \times \ln 2 = k_B \ln 2\left(\log_2\frac{V_i}{V_f}\right)$$

$$= (k_B \ln 2)\, I_{total}$$

So $\boxed{\Delta S = (k_B \ln 2)\, I_{total}}$

That's how change of entropy is related to the information.

(b) N spin-½ particle in external field B_0, at temp T.

The Hamiltonian of single particle $\rightarrow$

$$H = -g\frac{\mu_B}{2}(\vec{\sigma}\cdot\vec{B})$$

$$= -\mu B_0 \sigma_z$$ (assume magnetic field direction toward z-axis)

$$\mu = \frac{g\mu_B}{2}$$

Now, this hamiltonian has two eigenvalue which are $\pm \mu B_0$.

Single particle partition function Z_1

$$Z_1 = \sum_E e^{-\beta E} = e^{\beta\mu B_0} + e^{-\beta\mu B_0}$$

$$= 2\cosh(\beta\mu B_0)$$

Then N-particle partition function is $Z_N = Z_1^N$

$$Z_N = 2^N [\cosh(\beta\mu B_0)]^N$$

Free energy $\rightarrow$ $F = -k_B T \ln Z_N$

$$= -N k_B T [\ln 2 + \ln\cosh(\beta\mu B_0)]$$

Entropy → $S = -\frac{\partial F}{\partial T} = - k_B \ln Z$

$$= \frac{\partial}{\partial T}\left[N k_B T \ln 2 + N k_B T \ln \cosh\left(\frac{\mu B_0}{k_B T}\right)\right]$$

$$= N k_B \ln 2 + N k_B \ln \cosh\left(\frac{\mu B_0}{k_B T}\right) - \frac{N k_B T \sinh\left(\frac{\mu B_0}{k_B T}\right)}{\cosh\left(\frac{\mu B_0}{k_B T}\right)} \frac{1}{T^2} \left(\frac{\mu B_0}{k_B}\right)$$

$$= N k_B \left[\ln (2 \cosh(\beta \mu B_0)) - \frac{\mu B_0}{T} \tanh(\mu B_0 \beta)\right]$$

So Entropy →

$$N k_B \left[\ln (2 \cosh(\beta \mu B_0)) - \frac{\mu B_0}{T} \tanh(\beta \mu B_0)\right]$$

Q2. a) Toffoli Gate :- It flips the state of the third bit iff both the first and second bits are in state 1.

input

a	b	c
0	0	0
0	1	0
1	0	0
1	1	0
0	0	1
0	1	1
1	0	1
1	1	1

output

a	b	$c \oplus a.b$
0	0	0
0	1	0
1	0	0
1	1	1
0	0	1
0	1	1
1	0	1
1	1	0

(b) Note that, as per defination of Toffoli Gate, $G|110\rangle = |111\rangle$ and $G|111\rangle = |110\rangle$ others remains same. So, the matrix form will be

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

This block will do the change

c) $G = \begin{pmatrix} \mathbb{1}_{6\times 6} & 0 \\ 0 & \sigma_x \end{pmatrix}$

$$G^\dagger G = \begin{pmatrix} \mathbb{1}_{6\times 6} & 0 \\ 0 & \sigma_x \end{pmatrix}\begin{pmatrix} \mathbb{1}_{6\times 6} & 0 \\ 0 & \sigma_x \end{pmatrix} = \begin{pmatrix} \mathbb{1}_{6\times 6} & 0 \\ 0 & \mathbb{1}_{2\times 2} \end{pmatrix} = \mathbb{1}_{8\times 8}$$

So $\boxed{G^\dagger G = \mathbb{1}}$ Proved

3. a)

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} \cos\pi/4 & \sin\pi/4 \\ \sin\pi/4 & -\cos\pi/4 \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}\begin{pmatrix} \cancel{\cos} e^{i\pi/4}\cos\pi/4 & e^{i\pi/4}\sin\pi/4 \\ e^{-i\pi/4}\sin\pi/4 & -e^{-i\pi/4}\sin\pi/4 \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}\begin{pmatrix} e^{i\pi/2}\cos\pi/4 & \sin\pi/4 \\ \sin\pi/4 & -e^{-i\pi/2}\sin\pi/4 \end{pmatrix}\begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

$$= e^{i\pi/2}\begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}\begin{pmatrix} \cos\pi/4 & e^{-i\pi/2}\sin\pi/4 \\ e^{-i\pi/2}\sin\pi/4 & -e^{-i\pi}\sin\pi/4 \end{pmatrix}\begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4\cancel{-i\pi}} \end{pmatrix}$$ ~~$-i\sin(\pi)$~~ $+i\sin(-\pi)$

$$= e^{i\pi/2}\begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}\begin{pmatrix} \cos\pi/4 & e^{-i\pi/2}\sin\pi/4 \\ e^{-i\pi/2}\sin\pi/4 & \sin\pi/4 \end{pmatrix}\begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

$$R_z(\pi/2) = \cos\pi/4\,\mathbb{1} - i\sin\pi/4\,\sigma_z = \begin{pmatrix} \cos\pi/4 - i\sin\pi/4 & 0 \\ 0 & \cos\pi/4 + i\sin\pi/4 \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

$$R_x(\pi/2) = \cos\pi/4\,\mathbb{1} - i\sin\pi/4\,\sigma_x$$

$$= \begin{pmatrix} \cos\pi/4 & -i\sin\pi/4 \\ -i\sin\pi/4 & \cos\pi/4 \end{pmatrix} = \begin{pmatrix} \cos\pi/4 & e^{-i\pi/2}\sin\pi/4 \\ e^{-i\pi/2}\sin\pi/4 & \cos\pi/4 \end{pmatrix}$$

So, $\boxed{H = e^{i\pi/2}\,\cancel{R_z(\pi/4)}\,R_z(\pi/2)\,R_x(\pi/2)\,R_z(\pi/2)}$ → in terms of global phase & R_x, R_z.

(b)(i) $U = e^{i\alpha} R_{\hat{n}}$

The group of all 2×2 unitary is $U(2)$ which can be decomposed as

$U(2) = U(1) \times SU(2)$

which can be written as, $U = e^{i\alpha} V$ where $e^{i\alpha} \rightarrow$ global phase

$V \in SU(2)$

$\Downarrow$

$\det(V) = 1$

Hence V can be written in Pauli basis.

$$V = (a_0 \mathbb{I} - i\vec{a}\cdot\vec{\sigma})$$

$$V^\dagger V = (a_0 \mathbb{I} + i\vec{a}\cdot\vec{\sigma})(a_0\mathbb{I} - i\vec{a}\cdot\vec{\sigma})$$

$$= a_0^2 \mathbb{I} + i a_0 \vec{a}\cdot\vec{\sigma} - i a_0 \vec{a}\cdot\vec{\sigma} + |\vec{a}|^2 \sigma^2$$

$$= (a_0^2 + |\vec{a}|^2)\mathbb{I} \quad = \mathbb{I}$$

So, $(a_0^2 + |\vec{a}|^2) = 1$.

Let $a_0 = \cos\theta/2$; $|\vec{a}| = \sin\theta/2$ $\quad \hat{n} = \dfrac{\vec{a}}{|\vec{a}|}$

So,

$$V = \cos\theta/2\,\mathbb{I} - i\sin\theta/2\,(\hat{n}\cdot\vec{\sigma}) = R_{\hat{n}}(\theta)$$

So, $U = e^{i\alpha} R_{\hat{n}}(\theta)$ <u>Proved</u>

(ii) $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}(\sigma_x + \sigma_z)$

So, $\hat{n}\cdot\vec{\sigma} = \frac{1}{\sqrt{2}}(\sigma_x + \sigma_z)$

So, $\hat{n} = \frac{1}{\sqrt{2}}(1, 0, 1)$

& $H = e^{i\alpha} R_{\hat{n}}(\theta)$

$= e^{i\alpha}(\cos\theta/2 - i\sin\theta/2(\hat{n}\cdot\vec{\sigma}))$

$= \cancel{e^{i\alpha}(\cos\theta/2 - i\sin\theta/2\, e^{i\alpha}}$

But, $H = \hat{n}\cdot\vec{\sigma}$

So, $\cos\theta/2 = 0$ So, $\theta = \pi$.

$-i\sin\pi/2\, e^{i\alpha} = 1 \Rightarrow$ $e^{i(\alpha - \pi/2)} = e^0$

$\underline{\underline{\alpha = \pi/2}}$

So, $\theta = \pi$, $\alpha = \pi/2$

$H = e^{i\pi/2} R_{\hat{n}}(\pi)$ $\quad \hat{n} = \frac{1}{\sqrt{2}}(1, 0, 1)$

$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$

$= e^{i\pi/4} \begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$

$= e^{i\pi/4} \begin{pmatrix} \cos\pi/4 - i\sin\pi/4 & 0 \\ 0 & \cos\pi/4 + i\sin\pi/4 \end{pmatrix}$

$= e^{i\pi/4} \left(\cos\pi/4 \, \mathbb{I} - i\sin\pi/4 (\hat{n}\cdot\vec{\sigma}) \right)$ $\quad \hat{n} = (0,0,1)$

$\underline{S = e^{i\pi/4} R_z(\pi/2)}$

4. $(\vec{a}\cdot\vec{\sigma})(\vec{b}\cdot\vec{\sigma})$

$= \sum_{i,j} a_i b_j \sigma_i \sigma_j$

(we know, $\sigma_i \sigma_j = \delta_{ij} \mathbb{I} + i \sum_k \epsilon_{ijk} \sigma_k$

$= \sum_{i,j} a_i b_j (\delta_{ij} \mathbb{I} + i \sum_k \epsilon_{ijk} \sigma_k)$

$= \sum_{ij} a_i b_j \delta_{ij} \mathbb{I} + i \sum_{i,j,k} a_i b_j \epsilon_{ijk} \sigma_k$

$= \sum_i a_i b_i \mathbb{I} + i \sum_k (a \times b)_k \sigma_k$

$= (\vec{a}\cdot\vec{b}) \mathbb{I} + i \vec{\sigma}\cdot(\vec{a}\times\vec{b})$

So $\boxed{(\vec{a}\cdot\vec{\sigma})(\vec{b}\cdot\vec{\sigma}) = (\vec{a}\cdot\vec{b})\mathbb{I} + i\vec{\sigma}\cdot(\vec{a}\times\vec{b})}$ Proved